

SmritiCare NER: Cognitive Activities Clinical Guide

Scientific Rationale, Cognitive Domains & Therapeutic Benefits for Elderly Care

Cultural Focus: 8 North Eastern States | Target: Mild Cognitive Impairment & Dementia Care

1. CLINICAL PURPOSE & DESIGN PHILOSOPHY

SmritiCare NER provides culturally grounded, accessible cognitive stimulation designed to support memory preservation, executive functioning, and daily routine retention. Unlike anxiety-inducing clinical screening tools, every activity features stress-free pacing, multi-sensory audio guidance, and positive reinforcement.

Non-Diagnostic Safety: Zero failure screens or clinical labels. Emphasizes daily cognitive comfort & habit.

THE FOUNDATION: MEMORY BASELINE CHECK

PACE CALIBRATION

Why It Is Used:

A 2-minute, 5-task calibration completed upon registration to assess visual memory, sequence order, pair matching, short delayed recall, and pattern completion. Normalizes total performance to a 0-100 initial baseline score.

Adaptive Pacing Tiers Assigned:

± Gentle Support (0-59)

Max visual hints & timer allowances

Standard Rhythm (60-89)

Balanced pacing & cognitive cues

³ Active Challenge (90-100)

Brisk pace & independent sequencing

Patient & Caregiver Value:

Prevents task frustration by starting patients in their exact comfort zone. Retested anytime without clinical anxiety.

Note: Baseline completion does NOT increment daily streak, preserving integrity of real daily cognitive habits.

ACTIVITY 1: MEMORY WEAVE (Threads of My Life)

EPISODIC & SEMANTIC MEMORY

Cognitive Domain & Brain Area: Episodic Memory, Visual Pairing | Hippocampus & Temporal Lobe

Why It Is Used:

Stimulates visual-spatial memory and recall of hidden item locations using familiar North Eastern cultural motifs (Assam Tea Cup, Jaapi Hat, Muga Silk Shawl, Great Hornbill, Bamboo Craft, Bihu Dhol).

Adaptive Mechanics:

± Gentle: 2 pairs (4 cards), 3 hints allowed, 45s timer, full descriptive labels.

± Standard: 3 pairs (6 cards), 2 hints allowed, 35s timer.

± Active: 4 pairs (8 cards), 1 hint allowed, 25s brisk timer.

Therapeutic Benefit:

Taps into Reminiscence Therapy reserves. Cultural familiarity reduces cognitive resistance and reinforces synaptic pathways for visual retention and item location memory.

ACTIVITY 2: DAILY ROUTINE (My Day Flow)

EXECUTIVE FUNCTION & SEQUENCING

Cognitive Domain: Executive Function, Procedural Memory & Logical Order | Prefrontal Cortex

Why It Is Used:

Preserves the sequential organization of Activities of Daily Living (ADLs) such as morning wake-up, brewing tea, taking morning medications, tending the home garden, and evening rest.

Adaptive Mechanics:

- â€¢ Gentle: 3 core steps with numbered position helpers and audio hints.
- â€¢ Standard: 4 steps with drag-and-drop chronological sequencing.
- â€¢ Active: 5 multi-stage daily routine steps with nuanced afternoon/evening tasks.

Therapeutic Benefit:

Reinforces frontostriatal procedural memory pathways. Directly protects against domestic confusion and preserves functional independence in daily self-care routines.

ACTIVITY 3: SENSORY SOUNDSCAPE (Sounds of NER)

AUDITORY DISCRIMINATION & CALM

Cognitive Domain: Auditory Processing, Cross-Modal Association | Auditory Cortex & Limbic System

Why It Is Used:

Exercises sound-to-image association while lowering autonomic arousal, anxiety, and sundowning agitation through natural soundscapes (gentle rain on tin roof, Brahmaputra river breeze, mountain bird call, bamboo flute melody).

Adaptive Mechanics:

- â€¢ Gentle: 2 distinct choices (1 distractor), 4.0s sustained audio duration, unlimited audio replays.
- â€¢ Standard: 3 choices, 3.2s audio duration, unlimited replays.
- â€¢ Active: 4 choices with subtle acoustic differences (e.g. river stream vs temple waterfall).

Therapeutic Benefit:

Provides dual cognitive stimulation and therapeutic calming. Highly effective for agitated patients during evening hours.

ACTIVITY 4: RHYTHM WEAVER (Bihu & Folk Beats)

MOTOR TIMING & WORKING MEMORY

Cognitive Domain: Motor Timing, Audio-Motor Synchronization | Cerebellum & Premotor Cortex

Why It Is Used:

Engages musical working memory and fine motor tapping accuracy through traditional rhythmic drum patterns (Bihu Dhol, Dholak, Peepa, Khols).

Adaptive Mechanics:

- â€¢ Gentle: 3-beat rhythm pattern, relaxed 800ms tempo, generous 600ms tap timing tolerance.
- â€¢ Standard: 4-beat rhythm pattern, 650ms tempo, 450ms tap tolerance.
- â€¢ Active: 5-beat rhythm pattern, 500ms tempo, 350ms precise tap tolerance.

Therapeutic Benefit:

Rhythm engages intact musical memory networks, elevating mood, triggering positive dopamine release, and maintaining fine motor responsiveness on touchscreen devices.

ACTIVITY 5: DUAL-TASK JOURNEY (Focus & Stride)

DIVIDED ATTENTION & INHIBITION

Cognitive Domain: Divided Attention, Reaction Inhibition (Go/No-Go) | Frontoparietal Attention Network

Why It Is Used:

Trains the brain to handle two simultaneous cognitive tasks (monitoring stepping progress while reacting to auditory bell chimes). Divided attention decline is the strongest predictor of fall risk in elderly care.

Adaptive Mechanics:

â€¢ Gentle: 6 total steps, 2 chimes, generous 3200ms response window.

â€¢ Standard: 8 total steps, 3 chimes, 2200ms response window.

â€¢ Active: 10 steps, 4 chimes, 1400ms rapid response window.

Therapeutic Benefit:

Strengthens cognitive-motor dual-processing, reducing real-world tripping/fall incidents while walking and talking.

ACTIVITY 6: LIVING MARKET (Bazaar & Daily Essentials)

SEMANTIC MEMORY & BUDGETING

Cognitive Domain: Semantic Categorization, Working Memory & Mental Math | Prefrontal & Parietal Cortex

Why It Is Used:

Simulates shopping in a local North East vegetable and spice market (Ima Keithel / Fancy Bazaar). Patient memorizes and selects shopping list items (Fresh Ginger, Bamboo Shoots, Black Rice, Mustard Greens) while ignoring distractors.

Adaptive Mechanics:

â€¢ Gentle: 2 shopping items, 2 market distractors, visual list helper.

â€¢ Standard: 3 shopping items, 3 market distractors.

â€¢ Active: 4 shopping items, 4 distractors with basic price budgeting.

Therapeutic Benefit:

Preserves everyday functional competence in grocery selection, category discrimination, and domestic budgeting.

ACTIVITY 7: LANDMARK PATHFINDER (Heritage Trails)

SPATIAL ORIENTATION & NAVIGATION

Cognitive Domain: Spatial Navigation, Topographical Memory | Posterior Parietal & Entorhinal Cortex

Why It Is Used:

Reinforces spatial mental mapping and waypoint recognition across famous NE landmarks (Kamakhya Temple, Loktak Lake, Majuli Island, Tawang Monastery, Cherrapunji Root Bridges).

Adaptive Mechanics:

â€¢ Gentle: 2 landmarks on 3x3 grid with direct highlighted path guidance.

â€¢ Standard: 3 landmarks on 4x4 grid with step counter.

â€¢ Active: 4 landmarks across complex 4x4 route network.

Therapeutic Benefit:

Maintains environmental grid navigation cells, combating geographic disorientation and wandering tendencies.

ACTIVITY 8: COGNITIVE MINI-TEST (Daily Orientation)

TEMPORAL ORIENTATION & RECALL

Cognitive Domain: Temporal Orientation & Immediate Memory Recall | Hippocampus & Frontal Circuit

Why It Is Used: Provides a calm daily orientation reality check (day of week, current month, remembered flower, habit).

â€¢ Gentle: 3 questions with 2 large multiple-choice options.

â€¢ Standard: 4 questions with 3 choices.

â€¢ Active: 5 questions covering orientation, recall, and wellness habits.

Therapeutic Benefit: Anchors patients to time and calendar awareness daily in a supportive, dignity-first format.

ðŸŒŒ, EMOTIONAL REINFORCEMENT: MEMORY GARDEN

Why It Is Used: Converts cognitive effort into positive emotional reward instead of clinical percentage scores.

Mechanic: Every completed activity awards "Memory Dew Drops" to water virtual indigenous blossoms (Rhododendron, Lotus, Wild Orchid, Blue Vanda). Visual flower growth triggers positive dopaminergic habit loops.

ðŸŒ• CLINICAL & PRACTICAL VALUE FOR CAREGIVERS & HEALTHCARE

â€¢ Automated Telemetry: Family caregivers monitor longitudinal accuracy and streak habits without intrusive tests.

â€¢ Multi-Session Adaptive Rolling Engine: Evaluates 3-5 sessions before adjusting pace, eliminating single-day panic.

â€¢ Zero Diagnostic Burden: Frees caregivers from clinical scoring tension, fostering positive family engagement.

â€¢ Scalable for NER States: Fully operational across 8 regional languages with zero specialized medical hardware.

âšŸ ï• NON-DIAGNOSTIC CLINICAL SAFETY NOTICE & DISCLAIMER

1. SmritiCare NER activities, baseline check, and adaptive difficulty engine are designed exclusively for cognitive stimulation, daily routine maintenance, and non-diagnostic wellness engagement.
2. **This platform DOES NOT diagnose, treat, or cure Alzheimer's disease, dementia, or any clinical condition.**
3. Performance changes reflect activity comfort and should be reviewed by qualified medical professionals for any clinical diagnostic evaluation under MoHFW & MDoNER medical guidelines.